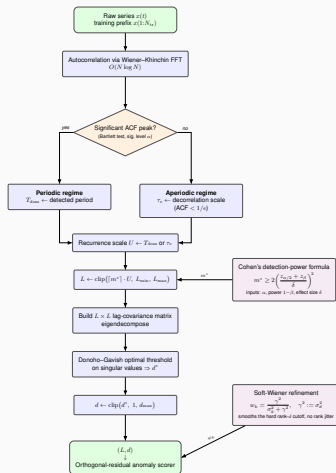


Closed-Form (L, d) Selection — Companion Notes

Full derivations and plain-English explanations, one slide per pipeline stage

Presenter's copy — not for the audience

Overview



Every box below gets its own slide, in this order.

Math

Series $x(t)$, healthy training prefix $x(1:N_{\text{tr}})$. A delay window of length L , stride τ :

$$w_t = [x(t), x(t + \tau), \dots, x(t + (L - 1)\tau)] \in \mathbb{R}^L.$$

From the training windows we will fit a d -dimensional healthy subspace

$U_d = [u_1, \dots, u_d] \subset \mathbb{R}^L$ (Step 7 onward). Everything downstream (the covariance matrix, the subspace, the anomaly score) is built from these window vectors. L and d have to be fixed *before* any of that can happen.

Intuition

Think of each window as a snapshot of L consecutive samples, stacked into a single point in $\mathbb{R}^L$. We are choosing two things: how wide a lens to look through (L), and how many independent directions of “normal variation” we’re willing to model (d). Too narrow a lens and you can’t see the shape of the recurring pattern at all. Too wide and (as we found empirically) the model gets *too* flexible — it starts being able to explain away anomalies as well as normal data.

Step 1 — Autocorrelation via Wiener–Khinchin i

Math

$$\rho(\ell) = \frac{1}{N \text{Var}(\mathbf{x})} \text{IFFT}(|\text{FFT}(\mathbf{x})|^2)(\ell)$$

The Wiener–Khinchin theorem says the power spectral density and the autocorrelation function are a Fourier transform pair. So instead of the naive $O(N^2)$ double loop over all lag pairs, we FFT the series, square the magnitude (the power spectrum), and inverse-FFT back — $O(N \log N)$. *Notation:* ℓ here is the ACF lag, swept over a whole range of values to trace out $\rho(\cdot)$ — deliberately distinct from τ on the previous slide, which is the (fixed, single-valued) delay-embedding stride. Same Greek letter would have meant two different roles for one symbol across two slides.

Step 1 — Autocorrelation via Wiener–Khinchin i

Intuition

Autocorrelation answers: “how similar is the series to a delayed copy of itself?” If you shift a sine wave by exactly one period, it lines up with itself perfectly (correlation = 1). Shift it by half a period and it’s anti-correlated. The autocorrelation function traces this out across every possible lag, and it’s the raw material for detecting periodicity in the next step. The FFT trick is purely computational — it doesn’t change what we’re computing, just how fast.

Step 2 — Is There a Real Period? i

Math

Bartlett's classical result: under the null hypothesis that the series is white noise, the sample autocorrelation at any lag k is approximately $\text{Normal}(0, 1/N)$, so its standard error is $\approx 1/\sqrt{N}$. The significance threshold at level α :

$$z_{\text{crit}} = \frac{\Phi^{-1}(1 - \alpha/2)}{\sqrt{N}}.$$

A candidate peak in $\rho(\ell)$ only counts as “real” if $\rho(\text{peak}) > z_{\text{crit}}$.

Step 2 — Is There a Real Period? i

Intuition

Any finite noisy series will show *some* bumps in its autocorrelation just by chance — the question is whether a given bump is big enough that it probably isn't just random luck. This is a hypothesis test: “could this correlation have arisen from pure white noise, given how much data we have?” More data (N larger) shrinks the threshold, so we can detect subtler, weaker periodicity with a longer healthy prefix. This test is what decides which branch we go down — periodic or aperiodic.

Math

Among all ACF peaks that cleared z_{crit} , take the one with the highest autocorrelation value:

$$T_{\text{dom}} = \arg \max_{\text{significant peaks } \ell} \rho(\ell).$$

Step 3a — Periodic Regime i

Intuition

This is the strongest, most reliable recurrence in the series — e.g. lag 24 for hourly-sampled data with a daily cycle, or lag 7 for daily-sampled data with a weekly cycle. T_{dom} becomes the “unit” that the window length L will later be built from (a whole number of these periods, not an arbitrary length).

Step 3b — Aperiodic Regime i

Math

No significant peak found. Fall back to the decorrelation scale:

$$\tau_e = \min\{\ell : \rho(\ell) < 1/e\}.$$

Step 3b — Aperiodic Regime i

Intuition

Even without a clean period, we still need *some* natural timescale to build a window from. τ_e is the lag at which the series has “forgotten” its past — how far back does today’s value still meaningfully influence tomorrow’s? The $1/e \approx 0.368$ threshold is the classic choice, directly analogous to a time constant in exponential decay. *Honest caveat:* later, we treat repeated decorrelation segments as if they were independent trials the same way we’d treat repeated periods — that’s a looser approximation here than in the periodic case, since decorrelation segments aren’t literal repeats of a pattern the way true periods are.

Step 4 — One Recurrence Scale i

Math

$$R \leftarrow T_{\text{dom}} \text{ (periodic)} \quad \text{or} \quad \tau_e \text{ (aperiodic)}.$$

Notation: R (recurrence scale) — deliberately not U , which is reserved for the SVD eigenvector/subspace basis used from Step 7 onward (as in the projector $U_d U_d^\top$). A scalar and a matrix sharing a letter is an easy way to confuse an audience mid-talk.

Step 4 — One Recurrence Scale i

Intuition

Rather than maintain two completely separate downstream formulas, both branches collapse into a single scalar R with the same operational meaning: “one natural repeat-unit of this process.” Everything from here on (the power-formula multiplier, the window-length rule) treats R identically regardless of which branch produced it.

Step 5 — Cohen's Detection-Power Formula i

Math

$$m^* \geq 2 \left(\frac{z_{\alpha/2} + z_{\beta}}{\delta} \right)^2$$

$z_{\alpha/2} = \Phi^{-1}(1 - \alpha/2)$ is the two-sided rejection threshold, set by the tolerated false-positive rate α . $z_{\beta} = \Phi^{-1}(1 - \beta)$ is how far into the shifted (true-anomaly) distribution the power requirement must reach. Averaging over m repeats shrinks the standard error by $\sqrt{m}$, giving $\delta\sqrt{m} \geq z_{\alpha/2} + z_{\beta}$; solving for m and doubling for the two-sample form gives the formula above. *Worked example* at this project's tuned defaults ($\alpha = 0.01$, power = 0.80, $\delta = 1.9$): $z_{\alpha/2} \approx 2.576$, $z_{\beta} \approx 0.842$, $m^* = 2(3.418/1.9)^2 \approx 6.48 \rightarrow 7$.

Step 5 — Cohen's Detection-Power Formula i

Intuition

If you only get one repeat of the healthy pattern to compare against, a subtle anomaly can easily hide inside ordinary noise. More repeats inside the window means you can average out the noise and tell “this cycle is genuinely different” from “this is just normal variability” with more confidence. Cohen's formula answers exactly *how many* repeats are needed for a stated confidence level (α , power) and a stated minimum anomaly size (δ). **Crucial honesty point:** δ , α , and power are assumptions we bring to the problem, not something the healthy training data can tell us — a healthy-only prefix contains no anomalies to measure δ from. This is why δ is one of only two constants in this whole pipeline that needs a small amount of labelled data to set (the other is $d_{\max}$, Step 9).

Step 6 — Window Length L

Math

$$L = \text{clip}(\lceil m^* \rceil \cdot R, L_{\min}, L_{\max})$$

L : window length (samples). m^* : required cycle count (Step 5). R : recurrence scale (Step 4). $L_{\min} = 2$: hard floor ($d \geq 1$ plus a nonzero residual dimension).
 $L_{\max} \approx N_{\text{tr}}/4$: stability ceiling.

Claim: SSA licenses $L = m \cdot R$ (integer multiple of the recurrence scale) for better separability of the periodic component. *Citation flagged:* previously attributed to “Golyandina & Zhigljavsky (2011)” – checked directly against the named journal issue and found not to exist. Treat as unverified pending a real source (full account: PROGRESS.md, "Citation integrity finding").

Step 6 — Window Length L_i

Intuition

Analogy: if you're averaging hourly temperature readings to characterize a daily cycle, averaging over exactly 24 hours captures one full cycle cleanly; averaging over 24.5 hours mixes in bits of two different times of day and blurs the signal. Whole-cycle windows keep the periodic structure's energy concentrated in a small, clean set of directions instead of smearing it across many. *We tested, not just trusted, this*: raising the window ceiling $L_{\max}$ from 256 to 512 — giving the multi-cycle rule more room to run — was tried and made every tested δ value *worse*, not better, on real data. Theory said bigger should be at least neutral; the data disagreed, and we reverted. Good example of “derive, then verify — don't just trust the theory.”

Step 7 — Healthy Subspace via Lag-Covariance i

Math

Stack all training windows as rows of $H_{\text{tr}} \in \mathbb{R}^{M \times L}$ (M = number of training windows — deliberately not K , to avoid sounding identical to the eigenvector index j below when spoken aloud). The lag-covariance matrix and its eigendecomposition:

$$C = \frac{1}{M} H_{\text{tr}}^{\top} H_{\text{tr}} \in \mathbb{R}^{L \times L}, \quad C u_j = \sigma_j^2 u_j, \quad s_i = \sqrt{M \sigma_i^2}.$$

This is the Karhunen–Loève decomposition of the training windows — equivalently, PCA on the windows, or the right singular vectors of H_{tr} . *Notation*: standardized on M throughout (code: `hankel_auto_ld.py:967`) — an earlier draft used K and M for the same quantity.

Step 7 — Healthy Subspace via Lag-Covariance i

Intuition

The eigenvectors are the principal directions of variation across all the healthy training windows. A signal that's mostly periodic or quasi-periodic concentrates almost all its energy into a small number of top eigenvectors (e.g. a pure sinusoid lives in an exactly 2-dimensional subspace — a sine/cosine quadrature pair). Pure noise, by contrast, spreads its energy roughly evenly across *all* L eigenvalues, with no preferred direction. Telling these apart — “how many of the top eigenvectors are real structure, versus noise?” — is exactly the rank-selection problem, solved in the next step.

Step 8 — Donoho–Gavish Optimal Threshold i

Math

$$\tau_* = \omega(\beta) \cdot \text{median}(s), \quad \beta = L/M, \quad d^* = \#\{i : s_i > \tau_*\}$$

$$\omega(\beta) \approx 0.56\beta^3 - 0.95\beta^2 + 1.82\beta + 1.43$$

Random matrix theory (Marchenko–Pastur) characterizes the singular-value distribution of a pure-noise matrix. Gavish & Donoho (2014, IEEE Trans. Info. Theory) derived $\omega(\beta)$, a closed-form function of only the matrix’s aspect ratio $\beta = L/M$, that gives the asymptotically MSE-optimal cutoff separating “signal” singular values from the noise bulk — *without ever needing to know the noise level in advance* (the median of the singular values itself serves as a robust noise-scale estimate).

Intuition

A smart, self-calibrating thresholding rule: instead of picking an arbitrary cutoff (“keep the top 5 eigenvectors”) or needing to know how noisy the sensor is beforehand, this formula adapts automatically to the shape of your specific data matrix and gives you the mathematically optimal place to draw the line.

Math

$$d = \text{clip}(d^*, 1, d_{\max})$$

$d_{\max}$ has no derivation the way m^* does — it was set by ablation on TSB-AD-U's labelled Tuning split, not asserted from theory. Currently $d_{\max} = 14$, which happens to tie the unclamped rank exactly under the project's current settings — a free safety margin, not something currently costing accuracy.

Step 9 — Capping the Rank i

Intuition

If d is left uncapped, the healthy subspace can have *so many* dimensions that it becomes flexible enough to reconstruct part of an anomaly's shape too — “expressive slack.” The anomaly then partially projects *into* the subspace instead of standing out as residual energy, and detection contrast collapses. This isn't hypothetical: one real dataset's VUS-PR (a detection-quality metric) collapsed from ~ 0.9 to 0.08 when the rank was left unclamped on it. $d_{\max}$ is the second (and last) constant in this whole pipeline that needs labelled data — everything else ($\mathcal{T}_{\text{dom}}$, τ_e , d^*) comes from the unlabelled healthy training data alone.

Result: $(L, d) \rightarrow$ the Scorer i

Math

$U_d = [u_1, \dots, u_d]$, the top- d eigenvectors from Step 7, define the healthy subspace. For any window w (training or test):

$$r_{\perp} = (I_L - U_d U_d^{\top}) w, \quad \|w\|_2^2 = \|U_d^{\top} w\|_2^2 + \|r_{\perp}\|_2^2.$$

$\|r_{\perp}\|_2$ is the anomaly score — the fraction of a window's energy the healthy subspace cannot explain (before the soft-Wiener refinement used throughout the rest of this project, which smooths the hard rank cutoff into a continuous weighting instead — next).

Result: $(L, d) \rightarrow$ the Scorer i

Math

In practice, the hard cutoff above is replaced by a continuous, soft-Wiener weighting of each eigendirection:

$$w_k = \frac{\gamma^2}{\sigma_k^2 + \gamma^2}, \quad \gamma^2 := \sigma_d^2.$$

σ_k^2 : k -th eigenvalue of C (Step 7), descending. γ^2 : the eigenvalue *at* the rank- d cutoff itself, not a separate parameter. Dominant direction ($\sigma_k^2 \gg \gamma^2$): $w_k \rightarrow 0$. Suppressed direction ($\sigma_k^2 \ll \gamma^2$): $w_k \rightarrow 1$. Removes the “rank jitter” a hard cutoff shows when two eigenvalues are close.

Result: $(L, d) \rightarrow$ the Scorer i

Math

Does the soft-Wiener refinement redefine the test window? No — w is untouched throughout; only the *projector* applied to it changes. The weights w_k above are assembled into a full projector over **all** L eigendirections (not just the ones past rank d):

$$P_{\perp}^{(\gamma)} = \sum_{k=1}^L w_k u_k u_k^{\top}, \quad r_{\perp}^{(\gamma)} = P_{\perp}^{(\gamma)} w.$$

Compare to the hard cutoff, $r_{\perp} = (I_L - U_d U_d^{\top})w$: that's the special case $w_k \in \{0, 1\}$ (a step function at rank d) instead of the continuous $w_k = \gamma^2 / (\sigma_k^2 + \gamma^2)$. Same input window w in both cases — the refinement is entirely in how the residual is *computed* from it, not in the window itself.

Result: $(L, d) \rightarrow$ the Scorer i

Intuition

Everything in this deck exists to answer one question honestly: *how wide a window, and how many directions of normal variation, should the scorer be allowed?* Get L too short or d too large and the method either can't see the recurring pattern, or becomes flexible enough to explain away real anomalies. This derivation replaces what used to be unexplained constants (a fixed $4\times$ multiplier, an inherited rank cap of 6) with numbers traceable to a specific, citable argument each.

Summary: What's Unsupervised, What Isn't i

Math

Quantity	Source
T_{dom}, τ_e	Healthy training data alone
d^* (Donoho–Gavish)	Healthy training data alone
δ (Step 5)	Labelled Tuning-split ablation
d_{max} (Step 9)	Labelled Tuning-split ablation

Summary: What's Unsupervised, What Isn't i

Intuition

The honest claim is “zero labels used on the series being evaluated,” not “zero labels used anywhere.” Two small global constants were tuned once, on a held-out labelled development set, then held fixed — exactly analogous to hyperparameter selection on a validation set in supervised learning. This is stated explicitly rather than folded into an unqualified “unsupervised” claim, because a reader should not come away believing the method needs no labelled data anywhere in its development.